Wenbin He

Ph.D. Student, Hunan University | Visiting Ph.D. Student, Singapore Management University

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Research Interests

Visual reasoning, Thinking with Images, and reinforcement learning. I am interested in how vision-language models learn to use visual tools to gather evidence during reasoning.

Education & Research Visit

Singapore Management University | Nov. 2026 - Present
Visiting Ph.D. Student, School of Computing and Information Systems
Host supervisor: Lizi Liao · Funded by the China Scholarship Council (CSC)

Hunan University | Sep. 2025 - Present
Ph.D. in Electronic Information · Advisor: Da Cao

Minzu University of China | Sep. 2022 - Jun. 2025
Master of Software Engineering

University of South China | Sep. 2018 - Jun. 2022
Bachelor of Software Engineering

Publications

Cross-Lingual Adaptation for Vision-Language Model via Multimodal Semantic Distillation
Yu Weng*, **Wenbin He***, Jun Dong, Chaomurilige, Xuan Liu, Zheng Liu
IEEE Transactions on Multimedia, 2025 · CCF-A

Zero-Shot Cross-Lingual Knowledge Transfer in VQA via Multimodal Distillation
Yu Weng*, Jun Dong*, **Wenbin He***, Chaomurilige, Xuan Liu, Zheng Liu, Honghao Gao
IEEE Transactions on Computational Social Systems, 2025

Enhancing Human-Computer Interaction Through Decoupling Motion and Camera Control in Human-Centric Video Generation
Guanfu Chen*, Xiao Liang*, **Wenbin He***
International Journal of Human-Computer Interaction, 2025 · CCF-B

LGVLM-mIoT: A Lightweight Generative Visual-Language Model for Multilingual IoT Applications
Yu Weng, Kunyu Yang, Zheng Liu, **Wenbin He**, Xiangyun Tang
IEEE Internet of Things Journal, 2025

Enhancing Text-Image Person Re-identification via Intra-Class Relevance Learning
Wenbin He, Yutong Gao, Wenjian Liu, Chaomurilige, Zheng Liu, Ao Guo
International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP 2024)

Enabling Trustable Financing: A Verifiable Privacy-Preserving Cross-Chain Protocol
Lidu Lou, **Wenbin He**, Xiangyun Tang, Yiying Yang, Tao Zhang, Yanyu Cheng, Cai Xu
IEEE/CIC International Conference on Communications in China Workshops (ICCC Workshops), 2024

* Equal contribution.

Research Experience

Hunan University · Doctoral Research | 2025 - Present
I study visual reasoning in vision-language models, with an emphasis on learning to use visual tools to inspect task-relevant image regions.

Aerospace Information Research Institute, Chinese Academy of Sciences | 2024 - 2025
As a research intern, studied architectures for high-resolution vision-language models and implemented data synthesis pipelines for counting, recognition, referring expressions, and visual grounding.

Minzu University of China · Master's Research | 2022 - 2025
Worked on cross-lingual adaptation for vision-language models and visual question answering using knowledge distillation. Also developed methods for text-image person re-identification and controllable human-centric video generation.

Selected Research Projects

Multimodal Digital Asset Generation | 2024 - 2025
Hainan Provincial Key R&D Project. Designed a framework for integrating data from multiple sources into a multimodal knowledge graph.

Trustable Cross-Chain System | 2022 - 2024
National Key R&D Program. Contributed to a privacy-preserving cross-chain protocol for supply-chain financing and led system testing and documentation.

AI-Aided Writing Robot System | 2022 - 2023
National Key R&D Program. Worked on using image captioning and visual question answering to align news imagery and text for summarization.

Patent Application

Yu Weng, Tianjiao Xing, Xiangyun Tang, **Wenbin He**, et al. *A Verifiable Privacy-Preserving Cross-Chain System Based on Relay Chain.* Published application CN117335955A.

Selected Awards

China Scholarship Council Scholarship for doctoral research visit, 2026.

Special Scholarship, Minzu University of China, 2023-2024.

National Third Prize, China University Service Outsourcing Innovation and Entrepreneurship Competition (team leader).

National Third Prize, 11th Lanqiao Cup (C/C++ National Finals).

National Third Prize, National College Student Software Testing Competition (Finals).

Provincial Third Prize, 16th Hunan Collegiate Computer Programming Contest (ACM Track).